Build your own website with GitHub Pages

A step-by-step guide for beginners

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1 What you will have at the end

By the end of this guide you will have a personal website at an address like `https://your-name.github.io`. It will have a home page, a page for your work and a CV page. It will work on phones and offer light and dark themes. It costs nothing to run, needs no server, and is yours to change.

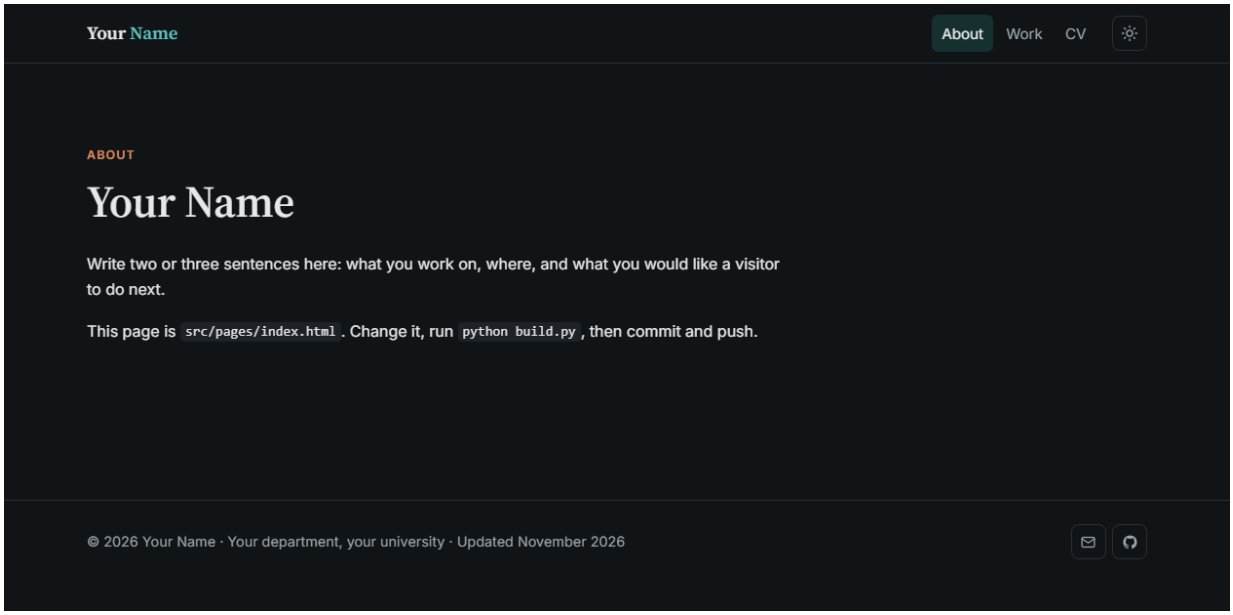


Figure 1.1: The site you will start from, before you change the words. You can open it at `ahmadammar.github.io/academic-site-template`.

1.1 Who this is for

You need no experience with code. If you can install a program and follow steps in order, you can do this. Set aside about an hour for the first time, most of it spent on installing things. After that, changing a page takes a few minutes.

Everything you need is in this guide, including in the printed copy. Web addresses are written in full so you can type them.

1.2 How it works, in one paragraph

GitHub offers a free service called GitHub Pages. It takes the files in a repository, which is a project folder stored on GitHub, and shows them as a website. So the job has four parts: get a copy of the template, change the words on your own computer, send the changes to GitHub, and switch Pages on.

1.3 The words you will meet

Git and GitHub have their own vocabulary. You do not need to memorise this table, so come back to it when a word looks unfamiliar.

Word	Meaning
Repository	A project folder whose history Git keeps. Often shortened to *repo*.
Commit	One saved version of your files, with a short note saying what changed.
Stage (or add)	Choosing which changes go into the next commit.
Push	Sending your new commits from your computer to GitHub.
Pull	Bringing new commits from GitHub to your computer.
Clone	Downloading a repository, with its history, to your computer.
Branch	A separate line of work, which lets you try a change without disturbing the main version.
`main`	The usual name of the main branch.
Fork	Your own copy, on GitHub, of someone else's repository.
Pull request	A request asking a project's maintainers to take in your changes.

1.4 The terminal, and how to read the commands here

A terminal is a window where you type instructions to your computer instead of clicking. It looks severe, but it only does what you type, and most commands in this guide are safe to repeat.

In VS Code, open the **Terminal** menu at the top of the window and choose **New Terminal**. A panel opens at the bottom of the window with a line that waits for you to type. That line is called the prompt. It usually ends in > or $ and shows the folder you are in.

When this guide shows a command, type everything on that line, then press Enter. Where a block shows a line beginning $, the $ stands for the prompt, so do not type it. The lines beneath it are what the computer prints back. Yours will match closely, although codes such as `1a2b3c4` will differ.

If a command prints something you did not expect, nothing is broken. Read the message, because it usually says what to do next. To stop a command that seems stuck, press Ctrl and C together.

2 Install the tools

You will install three free programs and create one account. Do these once and you will not need to repeat them.

2.1 Install VS Code, the program you will write in

VS Code is a free text editor from Microsoft. Go to `code.visualstudio.com`, press the large download button, open the file you receive, and accept the choices the installer offers. On Windows, leave the box marked **Add to PATH** ticked, because that lets the terminal find it.

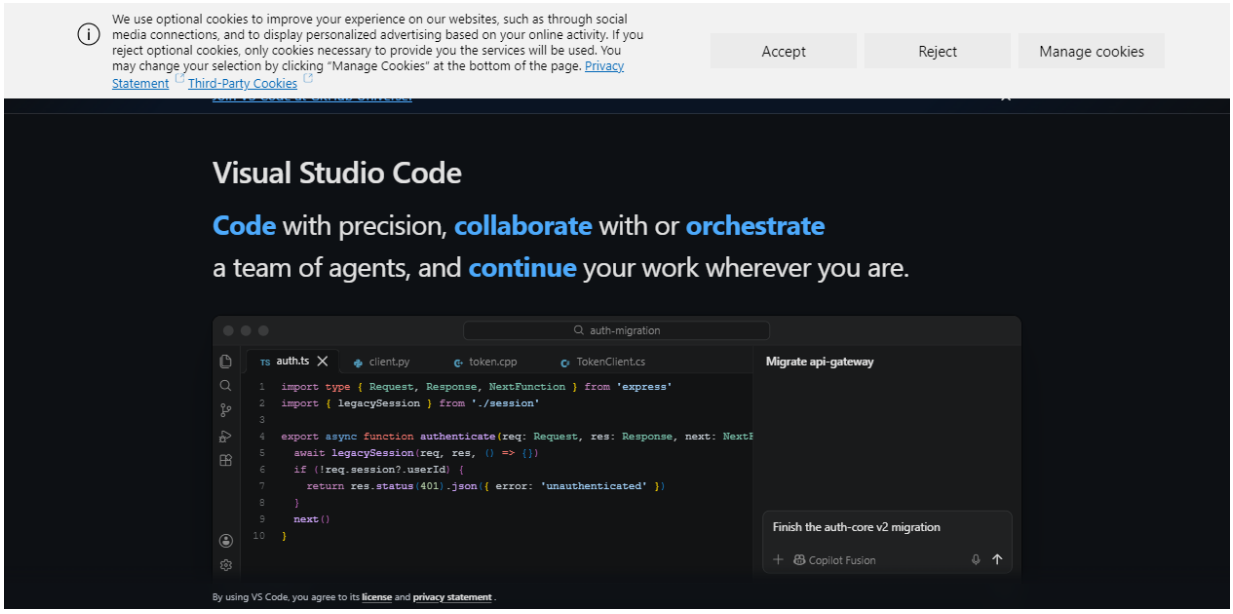


Figure 2.1: The VS Code download page. The site changes its wording from time to time, but the download button stays near the top. If a cookie notice appears, either choice is fine.

When VS Code first opens it may offer to sign you in to an assistant or to choose a theme. You can close these offers, since nothing in this guide needs them.

2.2 Install Git, the program that keeps your history

Git records each saved version of your files, so you can always go back. Go to `git-scm.com/downloads` and choose your operating system.

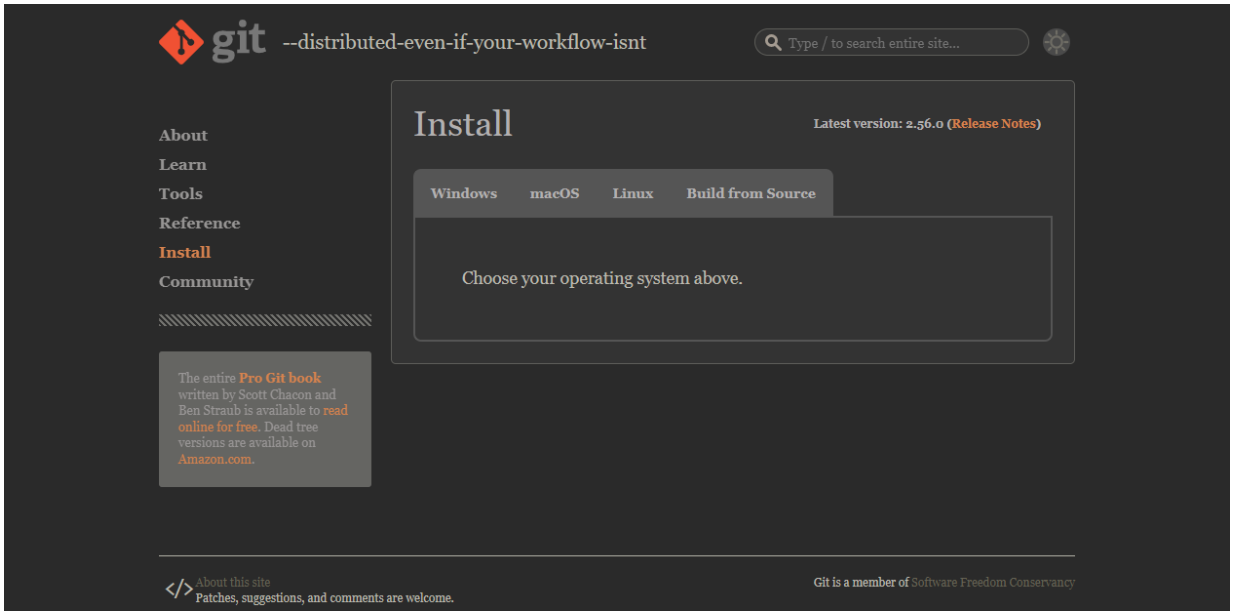


Figure 2.2: The Git download page. Choose Windows, macOS or Linux, then follow the instructions on the page that opens.

On Windows, run the installer and press **Next** on every screen, as the suggested settings are fine. On a Mac, open the Terminal application and type `git --version`, and the system will offer to install it. On Linux, use your system's package manager, for example `sudo apt install git`.

Close VS Code and open it again so that it notices Git. Then open a terminal and tell Git who you are. Git writes this name and email on every version you save, and they become public if you publish your work.

```
git config --global user.name "Your Name"
git config --global user.email "you@example.org"
git config --global init.defaultBranch main
```

These commands print nothing when they succeed. To check, type `git --version`, which should print a version number.

2.3 Install Python

Python is a programming language, and here it runs small scripts. Go to `python.org/downloads` and press the yellow download button.

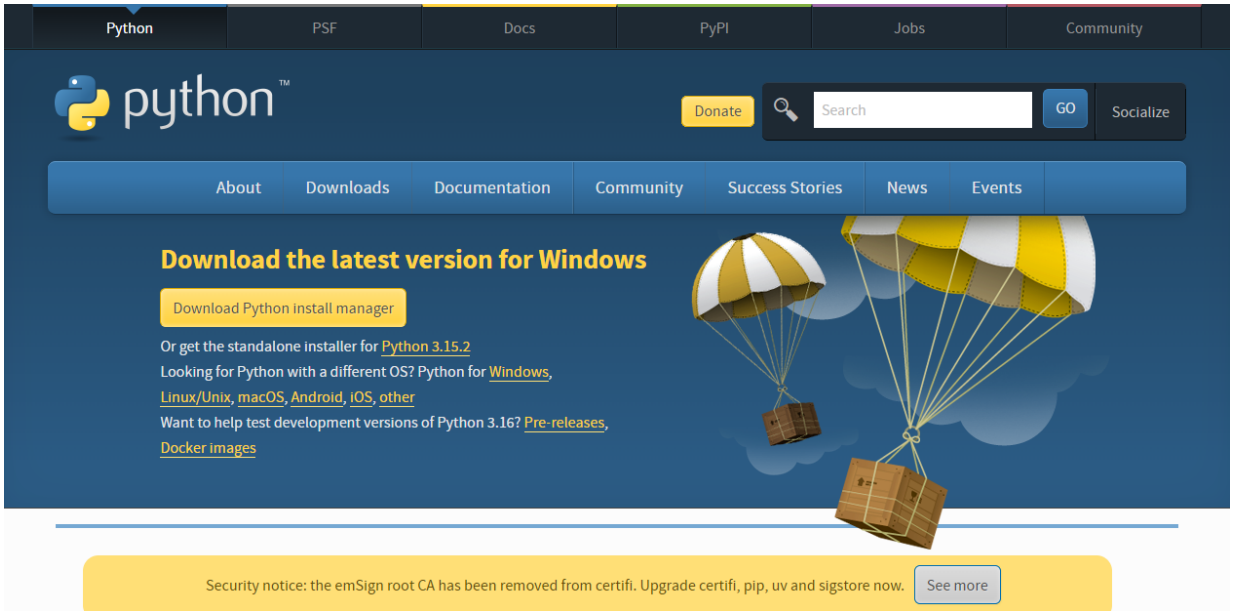


Figure 2.3: The Python download page. The page recognises your operating system and offers the right version.

On Windows, tick **Add python.exe to PATH** on the first screen of the installer before you press **Install Now**. It is easy to miss, and without it the terminal cannot find Python. Close and reopen VS Code, then check that it worked:

```
python --version
```

This should print a version number such as `Python 3.13.0`. On a Mac or on Linux, the command may be `python3` instead of `python`. Use whichever works, wherever this guide says `python`.

2.4 Create a GitHub account

GitHub is a website that stores copies of Git projects and lets other people see them. Git is the program on your computer, and GitHub is the place online.

Go to `github.com/signup`. Enter your email, choose a password and pick a username. Choose the username with care, because it appears in your web addresses and is awkward to change later. Your real name, or a close form of it, works well for professional use. GitHub will send a code to your email to confirm the account. The free plan is all you need, so if you are offered a paid plan or asked to fill in a survey, you can skip it.

The first time you send work to GitHub, a window will ask you to sign in through your browser. Accept it, and your computer will remember the permission.

3 Make your own copy of the site

3.1 Create the repository from the template

Sign in to GitHub, then go to `github.com/AhmadAmmar/academic-site-template`. Press the green button **Use this template**, then choose **Create a new repository**.

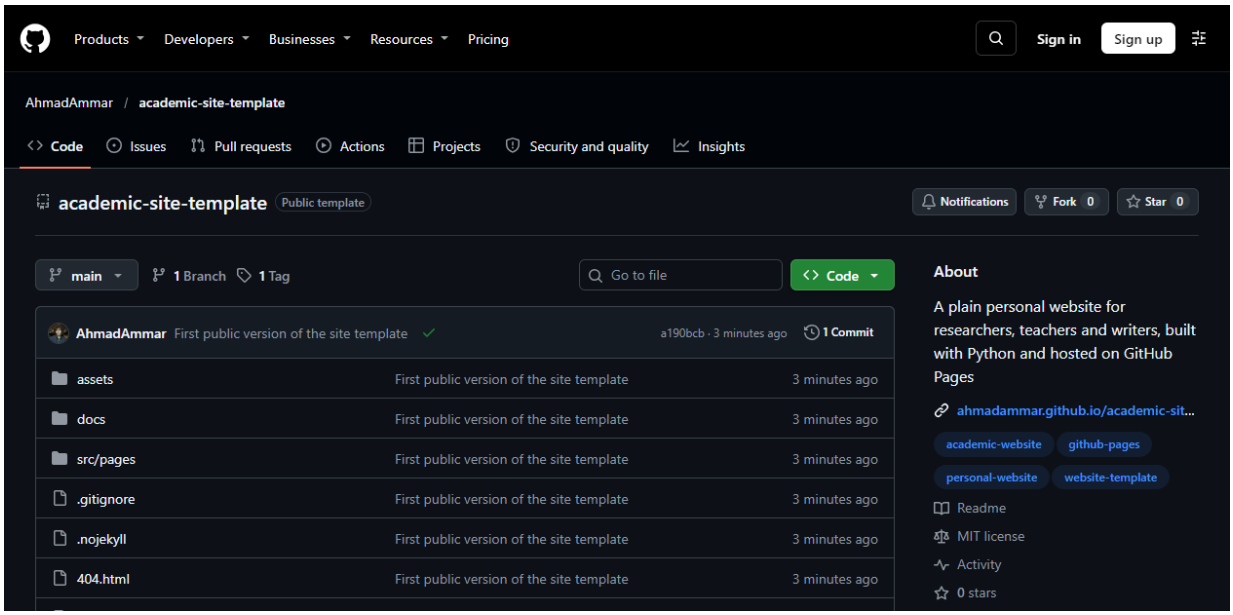


Figure 3.1: The template's page on GitHub. Once you are signed in, a green **Use this template** button appears near the top right, beside the list of files.

A form opens. Under **Repository name**, type your username followed by `.github.io`. If your username is `sara-khan`, the name is `sara-khan.github.io`. This exact form matters, because it is how GitHub knows the site belongs at your main address. Leave the repository set to **Public**, since free Pages sites need a public repository, then press **Create repository**.

3.2 Bring it to your computer

Your new repository is on GitHub, and now you need a copy on your computer. On its GitHub page, press the green **Code** button and copy the web address shown under **HTTPS**.

In VS Code, open the **Terminal** menu and choose **New Terminal**. Type these two commands, using your own address in the first one:

```
git clone https://github.com/your-name/your-name.github.io.git
code your-name.github.io
```

The first command downloads the project into a new folder. The second opens that folder in VS Code. If the second one fails, use the **File** menu, choose **Open Folder**, and pick the new folder.

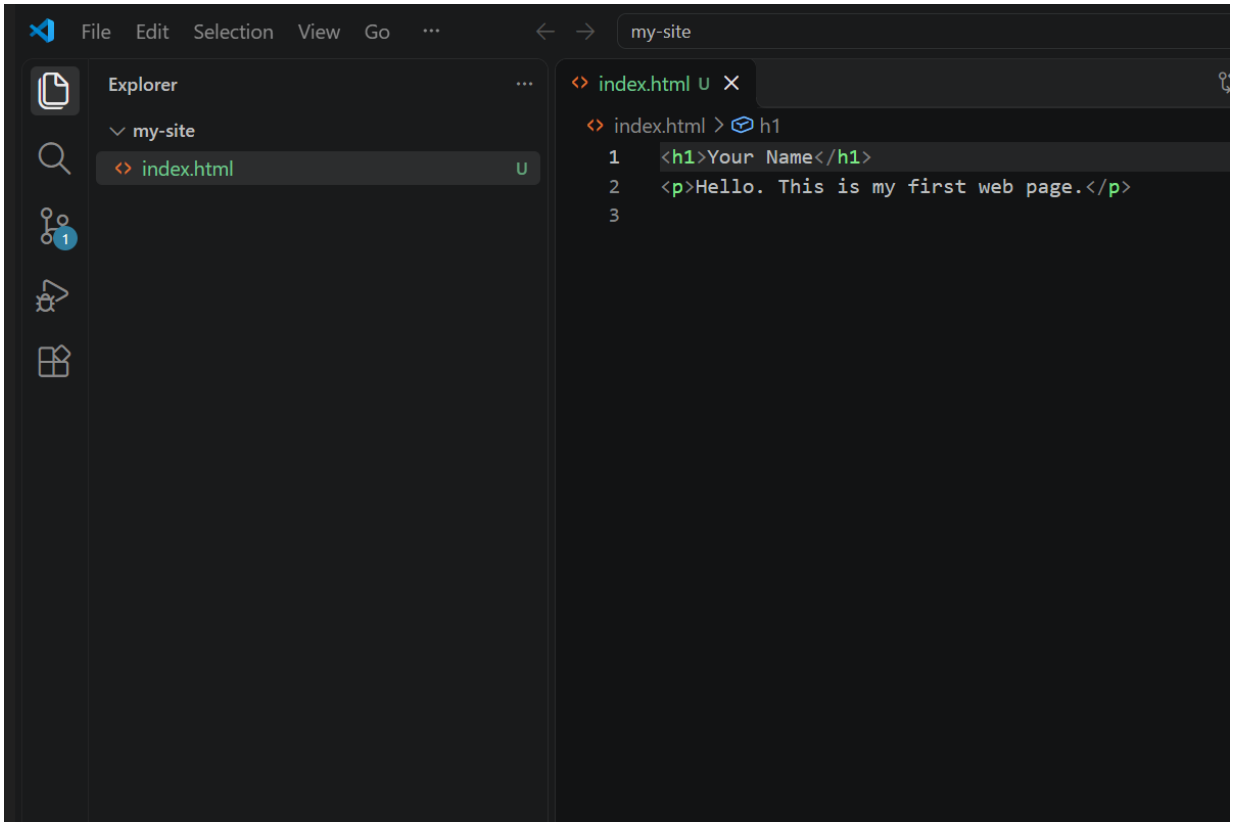


Figure 3.2: VS Code with a project open. The files are listed on the left, and the file you click opens on the right.

If VS Code asks whether you trust the authors of the files, say yes, since these are files you just chose to download.

3.3 What the files are

File or folder	What it holds
`site.toml`	Your name, address, menu and links. Start here.
`src/pages/`	The words of each page. You will edit these.
`assets/`	The look of the site. Leave it alone at first.
`build.py`	A small program that wraps your words in the layout.
`index.html`, `cv.html` and others	The finished pages, which `build.py` writes. Do not edit these by hand, because the next build replaces them.

4 Make it yours

4.1 Put your details in site.toml

Click `site.toml` in the list on the left. Change the words inside the quotation marks, and keep the quotation marks themselves. The `url` line must be your own address.

```
name = "Sara Khan"
url = "https://sara-khan.github.io"
affiliation = "Department of Geography, Example University"
```

Lower down, the `social` lines set the small icons at the bottom of each page. Change the email address and the GitHub address, and delete a whole line if you do not want that icon. Save with Ctrl and S, or Cmd and S on a Mac. A white dot on the file's tab means it has unsaved changes.

4.2 Write your home page

Open `src/pages/index.html`. The lines above the `---` describe the page, and everything below it is what visitors read. Text sits between tags, which are the words in angle brackets. `<h1>` marks the main heading, `<h2>` a smaller heading, and `<p>` a paragraph. A link looks like this:

```
<p>I study rivers at <a href="https://example.org">Example University</a>.</p>
```

Change the words and keep the tags. Then do the same in `work.html` and `cv.html`.

4.3 Build the site and look at it

The pages you edit are the raw material, so you now run the program that makes the finished site. In the terminal, type:

```
python build.py
python -m http.server 8000
```

The first command writes the finished pages. The second serves them on your own computer only. Open a browser and go to `http://localhost:8000`, and you will see your site. Nobody else can see this version.

When you have finished looking, click in the terminal and press Ctrl and C to stop the second command. Each time you change something, run `python build.py` again and refresh the browser.

4.4 Add a page, if you want one

To add a page about your teaching, make a new file, `src/pages/teaching.html`, that begins like this:

```
title: Teaching
description: Courses I teach.
```

```
nav: teaching
out: teaching.html
---
<section class="prose" style="padding:3rem 0">
  <h1>Teaching</h1>
  <p>Write about your courses here.</p>
</section>
```

Then add a line to the menu in `site.toml`, in the same form as the others:

```
{key = "teaching", label = "Teaching", href = "teaching.html"},
```

5 Publish it

5.1 Save a version and send it to GitHub

Publishing takes three commands. `git add` chooses what to save, `git commit` saves a version with a note, and `git push` sends it to GitHub. First, ask Git what has changed:

```
$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        index.html

nothing added to commit but untracked files present (use "git add" to track)
```

Your list will be longer, but it will have the same form. Files in red are changes that are not yet chosen. Now choose everything and save a version. The words in quotation marks are your note, so say what you did.

```
git add .
git commit -m "Add my details and first pages"
git push
```

The full stop after `add` means everything in this folder. Here is what the same steps printed for a small example with one file:

```
$ git add index.html
$ git commit -m "Add my first page"
[main (root-commit) cc3dbdc] Add my first page
 1 file changed, 1 insertion(+)
 create mode 100644 index.html
```

```
$ git push -u origin main
branch 'main' set up to track 'origin/main'.
To https://github.com/your-name/your-name.github.io.git
 * [new branch]      main -> main
```

The first time, a window will ask you to sign in to GitHub. Choose to sign in with your browser. If Git says *Please tell me who you are*, go back to the install chapter and run the two `git config` lines.

5.2 Switch on GitHub Pages

On your repository's GitHub page, press **Settings** in the row of tabs near the top. In the list on the left, press **Pages**. Under **Build and deployment**, set **Source** to **Deploy from a branch**. Under **Branch**, choose `main` and the folder `/ (root)`, then press **Save**.

5.3 See it online

GitHub now needs a minute or two to publish. To watch, press the **Actions** tab of your repository, where a yellow dot means it is working and a green tick means it has finished. Then open `https://your-name.github.io`. If you see an error page, wait two minutes and refresh, because a new site takes a moment to appear.

Your site is now public, and anyone with the address can read it.

5.4 The routine from now on

Every later change follows the same few steps, and this is all you need day to day:

```
python build.py
git add .
git commit -m "Say what changed"
git push
```

Each commit is kept, so you can see every earlier version in the repository's **Commits** page. Should you break something, the old version is still there.

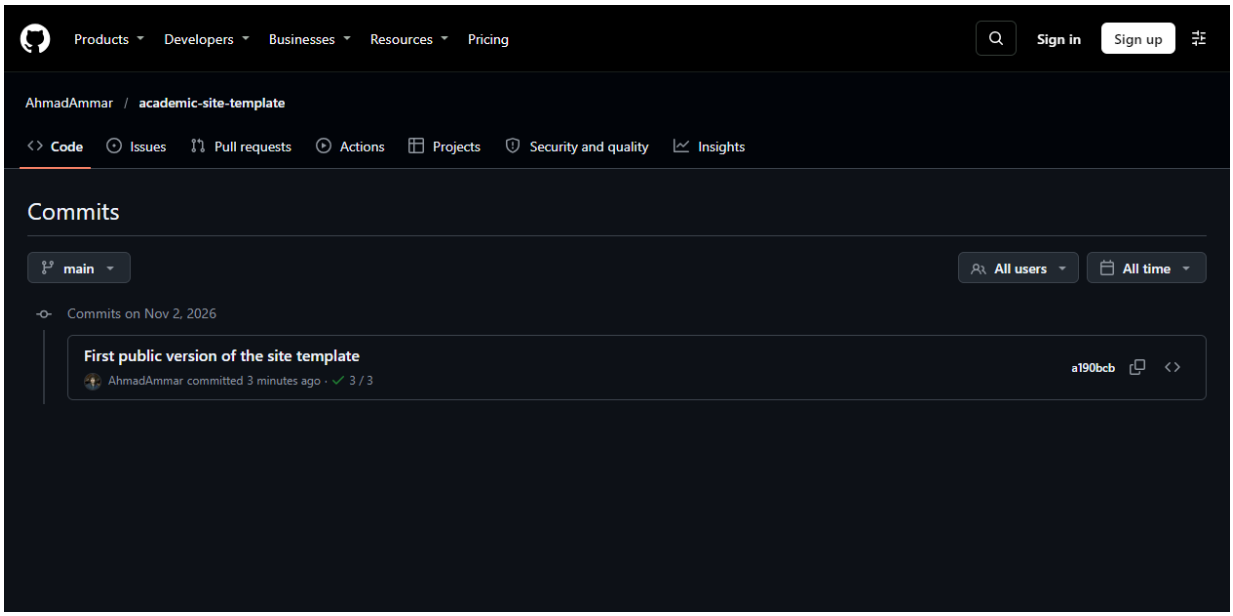


Figure 5.1: The list of saved versions on GitHub. Each entry shows its note and date, and can be opened to see exactly what changed.

6 When something goes wrong

What you see	What to do
`python` is not recognised	Python is missing from the PATH. Run the installer again, tick **Add python.exe to PATH**, then close and reopen VS Code. On a Mac or on Linux, try `python3`.
`git` is not recognised	Close and reopen VS Code. If that does not help, install Git again.
Please tell me who you are	Run the `git config` lines with your name and email, then repeat the commit.
nothing to commit	Git sees no changes. Check that the file is saved and that you ran `python build.py`.
failed to push and *fetch first*	GitHub has a version you do not, perhaps from an edit on the website. Run `git pull`, then `git push`.
The site shows a 404 page	Check the repository is named exactly `your-name.github.io`, that it is public, and that Pages is set to the `main` branch. Then wait two minutes.
The site looks unchanged	Your browser may be showing a saved copy. Hold Ctrl (Cmd on a Mac) and press R.
The page has no colours or layout	Check that the `url` line in `site.toml` is right and that the `assets` folder is still there.
A word in my page looks strange	An angle bracket or tag may be unclosed. Compare it with the sample tags in the chapter on making the site yours.

6.1 Taking a newer version of the theme

The template's look is kept apart from your words, and its versions are listed in `CHANGELOG.md` in the template repository. To take a newer theme, download the new `assets` folder and `build.py`, replace yours, and run `python build.py`. Your `site.toml` and `src/pages` stay as they are.